

1.9 Idempotent matrices

A square matrix $\mathbf{A}$ is **idempotent** if $\mathbf{A}^2 = \mathbf{A}$. Applying it twice has the same effect as applying it once, which is how a projection behaves. Examples are $\mathbf{I}$, $\mathbf{0}$, the centring matrix of Example (Sums of squares as quadratic forms), the products $\mathbf{AG}$ and $\mathbf{GA}$ for a generalized inverse $\mathbf{G}$ (Proposition 1.3.3(a)), and the non-symmetric $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$. In linear models, sums of squares are quadratic forms in *symmetric* idempotent matrices, and their degrees of freedom are ranks.

Proposition 1.9.1 — General idempotent matrices

Let $\mathbf{A}$ be $n \times n$ and idempotent of rank r .

- (a) Every eigenvalue of $\mathbf{A}$ is 0 or 1.
- (b) $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A})$.
- (c) $\mathbf{I} - \mathbf{A}$ is idempotent, $\mathbf{A}(\mathbf{I} - \mathbf{A}) = (\mathbf{I} - \mathbf{A})\mathbf{A} = \mathbf{0}$, $\mathcal{C}(\mathbf{I} - \mathbf{A}) = \mathcal{N}(\mathbf{A})$, and $\text{rank}(\mathbf{A}) + \text{rank}(\mathbf{I} - \mathbf{A}) = n$.
- (d) An idempotent matrix that is nonsingular equals $\mathbf{I}$.
- (e) $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ is idempotent for nonsingular $\mathbf{P}$, and $\mathbf{A}^\top$ is idempotent.

Proof.

- (a) If $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$, then $\lambda\mathbf{x} = \mathbf{A}\mathbf{x} = \mathbf{A}^2\mathbf{x} = \lambda^2\mathbf{x}$, so $\lambda^2 = \lambda$.
- (b) If $r = 0$ both sides vanish. Otherwise take a rank factorization $\mathbf{A} = \mathbf{B}\mathbf{F}$ (Theorem 1.2.1). Then $\mathbf{B}\mathbf{F}\mathbf{B}\mathbf{F} = \mathbf{B}\mathbf{F}$. Since $\mathbf{B}$ has full column rank and $\mathbf{F}$ has full row rank, $\mathbf{B}$ has a left inverse and $\mathbf{F}$ a right inverse (Section 1.3), and cancelling them gives $\mathbf{F}\mathbf{B} = \mathbf{I}_r$. So $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{B}\mathbf{F}) = \text{tr}(\mathbf{F}\mathbf{B}) = r$ by Theorem 1.6.1.
- (c) The first two claims are one-line expansions. $\mathcal{C}(\mathbf{I} - \mathbf{A}) \subseteq \mathcal{N}(\mathbf{A})$ because $\mathbf{A}(\mathbf{I} - \mathbf{A}) = \mathbf{0}$, and $\mathbf{v} \in \mathcal{N}(\mathbf{A})$ gives $\mathbf{v} = (\mathbf{I} - \mathbf{A})\mathbf{v}$. The rank statement is then rank-nullity.
- (d) Multiply $\mathbf{A}^2 = \mathbf{A}$ by $\mathbf{A}^{-1}$.
- (e) $(\mathbf{P}^{-1}\mathbf{A}\mathbf{P})^2 = \mathbf{P}^{-1}\mathbf{A}^2\mathbf{P}$ and $(\mathbf{A}^\top)^2 = (\mathbf{A}^2)^\top$.

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By (c), every $\mathbf{x}$ splits as $\mathbf{A}\mathbf{x} + (\mathbf{I} - \mathbf{A})\mathbf{x}$ with the first part in $\mathcal{C}(\mathbf{A})$ and the second in $\mathcal{N}(\mathbf{A})$. So $\mathbb{R}^n = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A})$, and $\mathbf{A}$ projects onto $\mathcal{C}(\mathbf{A})$ along $\mathcal{N}(\mathbf{A})$. The projection is orthogonal exactly when the two subspaces are orthogonal, which happens iff $\mathbf{A}$ is symmetric (Exercise (B1)).

Theorem 1.9.2 — Symmetric idempotent matrices

Let $\mathbf{A}$ be an $n \times n$ symmetric matrix of rank r . Then $\mathbf{A}$ is idempotent iff all its eigenvalues are 0 or 1. In that case

- (a) $\mathbf{A} = \mathbf{Q}_1\mathbf{Q}_1^\top$, where the $n \times r$ matrix $\mathbf{Q}_1$ has orthonormal columns spanning $\mathcal{C}(\mathbf{A})$;

- (b) $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}) = r$, the multiplicity of the eigenvalue 1;
- (c) $\mathbf{A}$ and $\mathbf{I} - \mathbf{A}$ are nonnegative definite, and $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \|\mathbf{A} \mathbf{x}\|^2 \leq \|\mathbf{x}\|^2$.

Proof. If $\mathbf{A}$ is idempotent its eigenvalues are 0 or 1 by Proposition 1.9.1(a). Conversely, if $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top$ with each $\lambda_i \in \{0, 1\}$, then $\mathbf{A}^2 = \mathbf{Q} \mathbf{\Lambda}^2 \mathbf{Q}^\top = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top = \mathbf{A}$. In that case, order the eigenvalues so that the r ones come first and let $\mathbf{Q}_1$ hold the corresponding eigenvectors. Then $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top = \mathbf{Q}_1 \mathbf{Q}_1^\top$, and $\mathcal{C}(\mathbf{A}) = \mathcal{C}(\mathbf{Q}_1)$ by Corollary 1.7.3(a). This is (a). The trace is the sum of the eigenvalues, which is (b). For (c), the eigenvalues of $\mathbf{A}$ and of $\mathbf{I} - \mathbf{A}$ are nonnegative, and $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x}$. Also $\|\mathbf{x}\|^2 - \|\mathbf{A} \mathbf{x}\|^2 = \mathbf{x}^\top (\mathbf{I} - \mathbf{A}) \mathbf{x} \geq 0$.

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Chapter 6 shows that symmetric idempotent matrices are exactly the orthogonal projections, and $\mathbf{Q}_1 \mathbf{Q}_1^\top$ in (a) is the projection onto $\mathcal{C}(\mathbf{A})$. In Chapter 4, part (b) becomes the degrees of freedom of a chi-squared distribution.

Theorem 1.9.3 — Sums of symmetric idempotent matrices

Let $\mathbf{A}_1, \dots, \mathbf{A}_k$ be symmetric $n \times n$ matrices and $\mathbf{A} = \sum_i \mathbf{A}_i$. Consider

- (i) $\mathbf{A}$ is idempotent;
- (ii) every $\mathbf{A}_i$ is idempotent;
- (iii) $\mathbf{A}_i \mathbf{A}_j = \mathbf{0}$ for all $i \neq j$.

Any two of these imply the third. Moreover, if $\mathbf{A} = \mathbf{I}_n$ and $\sum_i \text{rank}(\mathbf{A}_i) = n$, then (ii) and (iii) hold.

Proof.

(ii) and (iii) imply (i): $\mathbf{A}^2 = \sum_i \mathbf{A}_i^2 + \sum_{i \neq j} \mathbf{A}_i \mathbf{A}_j = \sum_i \mathbf{A}_i = \mathbf{A}$.

(iii) and (ii) imply (iii): By Theorem 1.9.2(c), $\|\mathbf{x}\|^2 \geq \mathbf{x}^\top \mathbf{A} \mathbf{x}$ for all $\mathbf{x}$. Fix j and take $\mathbf{x} = \mathbf{A}_j \mathbf{y}$, so that $\mathbf{A}_j \mathbf{x} = \mathbf{x}$ and $\mathbf{x}^\top \mathbf{A}_j \mathbf{x} = \|\mathbf{x}\|^2$. Then

$$\|\mathbf{x}\|^2 \geq \mathbf{x}^\top \mathbf{A} \mathbf{x} = \|\mathbf{x}\|^2 + \sum_{i \neq j} \mathbf{x}^\top \mathbf{A}_i \mathbf{x} = \|\mathbf{x}\|^2 + \sum_{i \neq j} \|\mathbf{A}_i \mathbf{x}\|^2.$$

So $\mathbf{A}_i \mathbf{A}_j \mathbf{y} = \mathbf{A}_i \mathbf{x} = \mathbf{0}$ for every $i \neq j$ and every $\mathbf{y}$.

(iv) and (iii) imply (ii): By (iii), $\mathbf{A} \mathbf{A}_i = \mathbf{A}_i^2$. Using $\mathbf{A}^2 = \mathbf{A}$, $\mathbf{A}_i^2 = \mathbf{A} \mathbf{A}_i = \mathbf{A}^2 \mathbf{A}_i = \mathbf{A} \mathbf{A}_i^2 = \mathbf{A}_i^3$. The eigenvalues of the symmetric $\mathbf{A}_i$ therefore satisfy $\lambda^2 = \lambda^3$, so they lie in $\{0, 1\}$, and $\mathbf{A}_i$ is idempotent by Theorem 1.9.2.

For the last statement, let $r_i = \text{rank}(\mathbf{A}_i)$. For each i , $\mathbf{I} - \mathbf{A}_i = \sum_{j \neq i} \mathbf{A}_j$ has rank at most $\sum_{j \neq i} r_j = n - r_i$ (Proposition 1.2.3(d)). So the eigenvalue 1 of $\mathbf{A}_i$, whose eigenspace is $\mathcal{N}(\mathbf{I} - \mathbf{A}_i)$, has multiplicity at least r_i . But $\mathbf{A}_i$ has exactly r_i nonzero eigenvalues, so all of them equal 1 and $\mathbf{A}_i$ is idempotent. Now (i) holds trivially, and (iii) follows from the second part of the proof.

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The last statement is the algebraic core of Cochran's theorem ([Chapter 4](#)): if $\mathbf{y}^\top \mathbf{y} = \sum_i \mathbf{y}^\top \mathbf{A}_i \mathbf{y}$ and the ranks add up to n , the quadratic forms are built from mutually orthogonal projections.

1.9.1 Exercises

1.9.1.1 A. Check your understanding

Exercise A1.

Let $\mathbf{A}, \mathbf{B}$ be idempotent of the same size. Show that $\mathbf{AB}$ is idempotent if $\mathbf{AB} = \mathbf{BA}$, and that $\mathbf{A} + \mathbf{B}$ is idempotent iff $\mathbf{AB} + \mathbf{BA} = \mathbf{0}$. Show that the latter forces $\mathbf{AB} = \mathbf{0}$ (symmetry is not needed).

1.9.1.2 B. Practice

Exercise B1.

Let $\mathbf{A}$ be idempotent. Show that $\mathbf{A}$ is symmetric iff $\mathcal{C}(\mathbf{A}) \perp \mathcal{N}(\mathbf{A})$.

Solution. Suppose $\mathbf{A}$ is symmetric. For $\mathbf{u} = \mathbf{Ax} \in \mathcal{C}(\mathbf{A})$ and $\mathbf{v} \in \mathcal{N}(\mathbf{A})$, $\mathbf{u}^\top \mathbf{v} = \mathbf{x}^\top \mathbf{Av} = 0$. Conversely, suppose $\mathcal{C}(\mathbf{A}) \perp \mathcal{N}(\mathbf{A})$. For all $\mathbf{x}, \mathbf{y}$, $(\mathbf{I} - \mathbf{A})\mathbf{y} \in \mathcal{N}(\mathbf{A})$, so $\mathbf{x}^\top \mathbf{A}^\top (\mathbf{I} - \mathbf{A})\mathbf{y} = 0$. Hence $\mathbf{A}^\top (\mathbf{I} - \mathbf{A}) = \mathbf{0}$, that is, $\mathbf{A}^\top = \mathbf{A}^\top \mathbf{A}$. The right side is symmetric, so $\mathbf{A}^\top$ is symmetric, and so is $\mathbf{A}$.